

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Science in Information Technology (B.Sc. IT)

Semester-I University Examination March-2018

CA 403 Information Technology and Digital Electronics

Date: 27.03.18, Tuesday Time: 10.00 a.m. to 01.00 p.m. Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.
4. Simple Calculator is permitted.

SECTION-1

Q.1 [A] Match column A with the correct answer on column B. [05]

Column A

1. SRAM
2. Archival Store
3. Main Memory
4. Interrupt
5. NIC

Column B

- a. Signal
- b. Volatile
- c. Cache Memory
- d. Connect to Communication Network
- e. Long Life

[B] Give an appropriate answer to the following multi choice questions. [06]

1. _____ is not an Input unit of a computer.
(a) Video Camera (b) NIC (c) Microphone (d) Scanner
2. _____ processor is used to process the audio and video data.
(a) Digital Signal Processor (b) Control Processor
(c) Super Processor (d) Embedded Processor
3. The data and instructions in _____ are stored by the manufacturer at the time of its manufacturing.
(a) DRAM (b) SRAM (c) ROM (d) Hard Disk
4. Adaptive security architecture is having _____ security measures to protect organization information.
(a) Strict (b) Rigid (c) Flexible (d) None of these
5. The area required to fabricate capacitor storage is _____.
(a) 0.01cm^2 (b) $0.02\mu\text{m}^2$ (c) 0.02cm^2 (d) $0.01\mu\text{m}^2$
6. The literary work is _____ type of intellectual property.
(a) Industrial (b) Copyright (c) Trademark (d) Trade secrets

Q.2 [A] What do you mean by Information Technology? Differentiates Data and Information. Give different types of Data with suitable examples. [06]

[B] Give characteristics of storage cell and explain Magnetic Storage Cell in details. [06]

OR

Q.2 [A] Give societal benefits of using WWW(at least twelve). [06]

[B] Explain Bluetooth and Infrared Communication in details. [06]

Q.3 Answer following questions in detail (Any Three). [12]

- 1.Explain Industry 4.0 and Bigdata Analytics concepts.
- 2.Differentiate RAM and ROM.

[P.T.O]

3. Give the steps of data processing using computer.
4. Differentiate Static RAM and Dynamic RAM.
5. Explain Capacitor storage cell.

SECTION-2

Q.4[A] Match column A with the correct answer on column B. [05]

Column A

1. NOT gate
2. Combinational Circuit
3. Sequential Circuit
4. Encoder
5. Register

Column B

- a. No Memory
- b. Binary Information
- c. Inverter
- d. Flip-flop
- e. Inverse of Decoder

[B] Give an appropriate answer from following multi choice questions. [06]

1. What is the decimal value of octal number 125?
(a) 84 (b) 85 (c) 75 (d) 74
2. The _____ gate is an electronic circuit that gives a high output only if all its inputs are high.
(a) OR (b) NAND (c) AND (d) None of these
3. 10's complement of 135 is _____.
(a) 864 (b) 865 (c) 765 (d) None of these
4. _____ is not a valid octal number.
(a) 125 (b) 138 (c) (a) and (b) both (d) None of these
5. What is the hexadecimal value of 109 Decimal number?
(a) 6E (b) 6D (c) 5D (d) 5E
6. What is the binary conversation of 0.625 decimal number?
(a) 0.100 (b) 0.110 (c) 0.101 (d) 0.111

Q.5 [A] Convert the following decimal numbers to equivalent binary numbers : [06]
(i) 998 (ii) 556 (iii) 778

[B] Explain AND, NOT and EXOR gates in details. [06]

OR

Q.5 [A] Perform following subtraction using 9's complements : [06]
(i) 556 – 230 (ii) 198 – 780 (iii) 350 - 546

[B] What do you mean by flip flop? What are the common basic characteristics of all kinds of flip flop? Explain S-R and Master-Slave flip flop in details. [06]

Q.6 Answer in detail (Any Three). [12]

1. Perform the following binary addition:
(a) 00011010 + 00001100 (b) 00010011 + 00111110
2. Explain NAND and NOR gate with truth table.
3. Draw the structure of Karnaugh Map based on two, three and four variables.
4. Explain Multiplexer in details.
5. Simplify the Boolean algebra $(x + y)(x + z)$ using Boolean algebra rules.